

Generation Interconnection System Impact Study Report

For Queue Project Q21-1968

Hubbardstown 138 kV

December 31, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1968. The Interconnection Customer proposes a 47.0 MW solar photovoltaic facility interconnecting at the Hubbardstown 138 kV substation in the Kentucky area. The request was received on December 28, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 684 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$275,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
TRISTATE 345/1 kV Auto	HUBBARDSTOWN – TISTATE 345kV	100.7	623	12.2
TRISTATE 138-1 kV XFMR	HUBBARDSTOWN – TRISTATE 345kV	100.7	623	12.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
TRISTATE 345/1kV Auto	Replace transformer with higher-rated unit	\$22,244,000	1% (\$234,000)
TRISTATE 138-1 kV XFMR	Replace transformer with higher-rated unit	\$3,929,000	1% (\$41,000)
	Total allocated to project		\$275,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.